

Imtiaz Ahmed

imtiaz92407@gmail.com • (917) 939-2234

EDUCATION

The Cooper Union for the Advancement of Science and Art

2029

B.E. – Mechanical Engineering

GPA: 3.975/4.000

Relevant course work: Eng. Design and Problem Solving, Principles of Design, Intro to Computer Science

Scholarships: Half-Tuition Scholarship, Innovator Scholarship

RELEVANT EXPERIENCE

Ecolibrium at Loisaída, Manhattan, NY

January 2026 – Present

Engineering Intern

- Designed air quality sensor enclosure to improve ESP32 and sensor integration
- Optimized layout for cable and antenna routing while isolating electronic components from interference
- Iterated design for manufacturability and designed hinged jig to facilitate assembly

Cooper Union Satellite Initiative, Manhattan, NY

September 2025 - Present

ACDS Subteam Member

- Designed rig for rigorous testing of magnetorquer functionality in one dimension
- Validated magnetorquer calculations by measuring resistivity
- Designed mounting brackets to securely hold PCBs in satellite frame

Cooper Union Bio-inspired Robotics Design Lab, Manhattan, NY

February 2026 - Present

Research Assistant

- Documented flapping wing manufacturing process with PLA and plastic membrane
- Designed hand-operated friction-based membrane tensioning system
- Conducted rapid prototyping to determine optimal membrane placement

Cooper Air Design-Build-Vertical Flight Team, Manhattan, NY

January 2026 - Present

Mechanical Subteam Member

- Designed LiDAR mounting system minimizing structural modifications to UAV frame plates
- Incorporated tool access constraints to allow efficient installation
- Designed battery mount to cleanly interface with power distribution board and existing attachment points

PROJECTS

Tendon-actuated Robotic Arm

- Designed and printed a fully 3D-printable Arduino-controlled robotic arm in OnShape
- Used linkage-based actuation of shoulder, elbow, and claw joints to keep all servos near arm base
- Moved center of gravity 40% closer to the base at maximum extension compared to a similar arm without linkages, a difference that would increase even more with heavier servos

VEX Ball Launching Robot

- Designed a robot using a system of conveyor belts and a ratcheting puncher mechanism to pick up balls from the ground and launch them into a basket
- Used OnShape to assemble robot using models of existing VEX pieces and custom-designed parts
- Coded controls for teleoperation in RobotC

SKILLS

CAD: AutoCAD; OnShape; Fusion 360

Data: MATLAB; Excel; Microsoft Office

Programming languages: C; Python